

An Alternative Proof of an Elementary Property of Pythagorean Triples

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1 Introduction

When I was a freshman in college I was briefly interested in the idea of using the areas of geometric figures to produce interesting formulas. The example that inspired me was the way one may derive the Pythagorean formula by examining a dissected square composed of an inscribed square and triangles.

I tried to do something similar with a more complicated geometric figure and I came up with figure 1.

By inscribing a circle¹ in a *right* triangle, one can decompose the triangle into two pairs of right triangles and a square. The fact that the sub-triangles are right triangles follows using the fact that any radial segment of a circle is perpendicular to its associated tangent line.

From the figure, the lower right four sided sub-figure has equal sides, r . I claim that this sub-figure is a square and not a rhombus. This can be seen as three of its interior angles are 90° ; and, as the interior angles of a four-sided *convex* polygon must sum to 360° , the last interior angle must also be 90° . Consequently, the four-sided figure is a square.

¹Existence and uniqueness of the inscribed circle in a right triangle is a standard fact: the angle bisectors are concurrent at the incenter, which lies at distance $r = (a + b - c)/2$ from each side. With the right-angle vertex at $(a, 0)$ and the legs along $y = 0$ and $x = a$, the incenter has coordinates $(a - r, r)$.

Below we find a well known relation between the sides, a , b , and c of the original triangle when the sides are integers in *reduced form*.² This is done by investigating the fact that the sum of the areas of these component triangles and square must be the same as the area of the original triangle.

2 A Basic Property of Primitive Pythagorean Triples

Writing the equivalence of the area of the figure and its parts we have:

$$\overbrace{r^2}^{\text{Square}} + \overbrace{2 \frac{r(a-r)}{2}}^{\text{Two lower triangles}} + \overbrace{2 \frac{r(b-r)}{2}}^{\text{Two upper triangles}} = \overbrace{\frac{ab}{2}}^{\text{Original triangle}} \quad (1)$$

Simplifying, this is:

$$r^2 + r(a-r) + r(b-r) = \frac{ab}{2} \quad (2)$$

One can solve the quadratic for r (the quadratic formula gives $r = (a+b \pm c)/2$; only the minus sign is geometrically valid, since $r < a$ and $r < b$) or read off from the figure that $(a-r) + (b-r) = c$. Solving this last equation for r gives:

$$r = \frac{a+b-c}{2} \quad (3)$$

A *primitive Pythagorean triple* (a, b, c) is a triple of positive integers in reduced form satisfying $a^2 + b^2 = c^2$; it gives the side lengths of a *primitive Pythagorean triangle*, where c is the hypotenuse and a, b are the legs. What can we say about the evenness or oddness of the sides when they have no common factor? The possibilities are:

- c is even AND *exactly* one of the following is true:

- Both a and b are even;

²A triple of integers, (a, b, c) is in reduced form if they have no common integer divisor.

- Both a and b are odd;
- c is odd AND *exactly* one of the following is true:
 - Exactly one of a and b is even (the other being odd).

Since the triple (a, b, c) is in reduced form, we can eliminate the case of all three being even. By the Pythagorean theorem we know that $a^2 + b^2 = c^2$. Using this we can reduce the combinations further to the following:³

- c is even AND both a and b are odd;
- c is odd AND exactly one of a and b is even (the other being odd).

Regardless of which choice occurs, equation (3) implies that the radius of the circle, r , is an integer. Using this fact and equation (2) we see that $\frac{ab}{2}$ must be an integer, which means the product ab is even. This knowledge eliminates the first choice; namely, that c is even with both a and b odd. Consequently, c is odd, with exactly one of a and b even (the other being odd). Without loss of generality we assume that b is even; in which case, a is odd.

Now, the left hand side of (2) simplifies to $r(a + b - r) = r(r + c)$ by (3); with c odd, this is even regardless of r 's parity. This implies that $\frac{ab}{2}$ is even. Since b is even, we can write this fraction as: $a\frac{b}{2}$. So we know that $a\frac{b}{2}$ (the product of two integers) is even. And we know that a is odd; therefore, $\frac{b}{2}$ is even. This means that b is divisible by 4.

In summary, we have deduced the following properties for primitive Pythagorean triangles.

- One of the legs is odd; the other leg is even; and the hypotenuse is odd.
- The even leg is divisible by 4.

³We list the results of all combinations of even and odd numbers with respect to: addition, subtraction, and multiplication:

1. $E \pm E = E$; $E \times E = E$;
2. $E \pm O = O$; $E \times O = E$;
3. $O \pm O = E$; $O \times O = O$.

These equations are to be read: An even integer plus (or minus) an even integer is an even integer. Likewise, an even integer plus (or minus) an odd integer is an odd integer, etc. These results are used throughout the rest of the document.

- The radius of the inscribed circle in the Pythagorean triangle is an integer (necessarily positive).

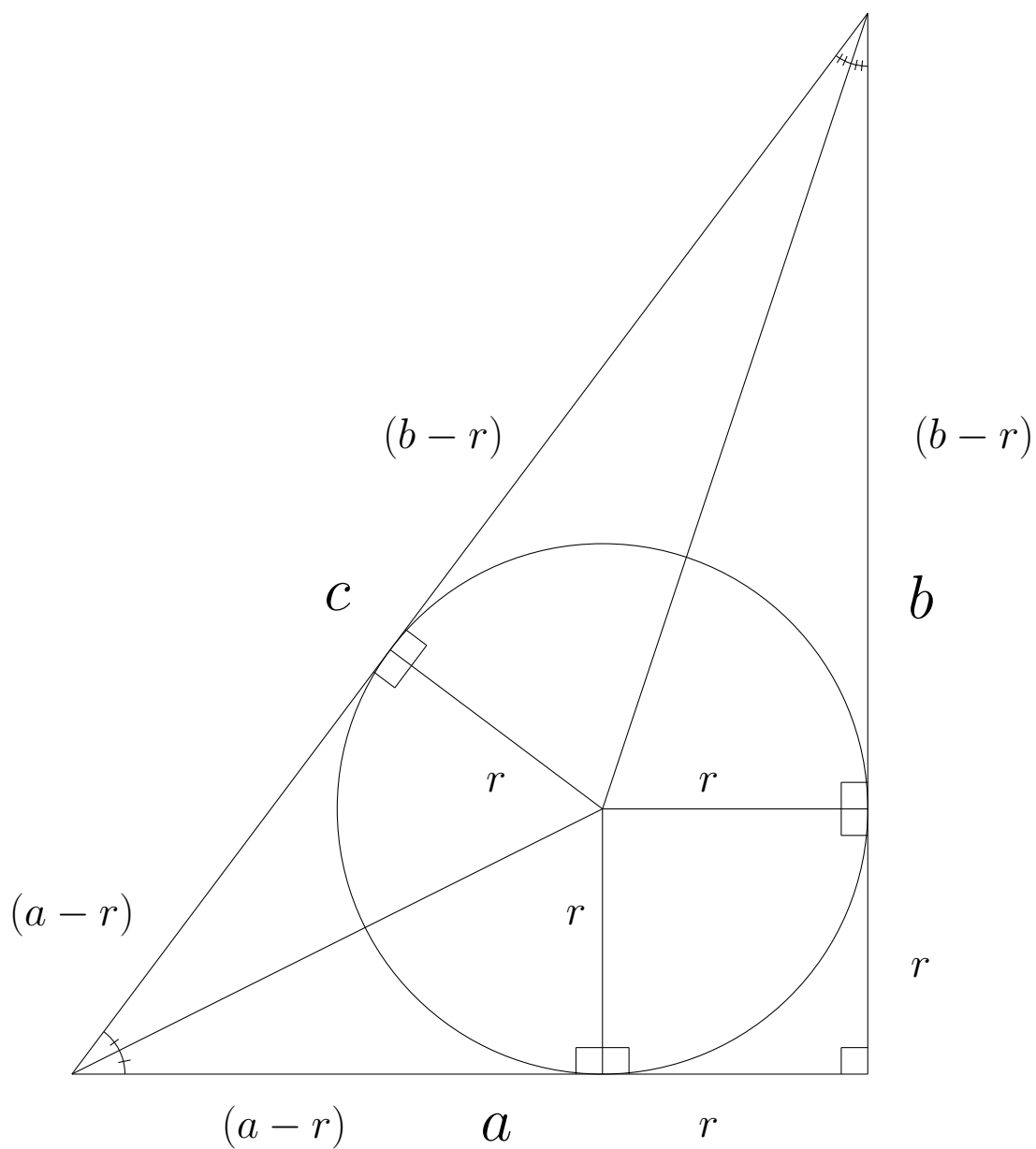


Figure 1: Dissected Right Triangle